



Association between exposure to polycyclic aromatic hydrocarbons and attention deficit hyperactivity disorder in children: a systematic review and meta-analysis

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Abstract

Some studies have shown that exposure to polycyclic aromatic hydrocarbons (PAHs) is a dangerous factor for attention deficit hyperactivity disorder (ADHD). This systematic review and meta-analysis aimed to clarify this relationship, and to collect and analyze all the relevant evidences in published reports of epidemiologic studies. PubMed, Science Direct, Web of Science, Scopus, and Google Scholar databases were searched through September 31, 2018. The study quality was evaluated using the Newcastle-Ottawa Scale. Moreover, fixed- and random-effect models were used. The data in this meta-analysis were presented as adjusted odds ratio (AOR). From 959 articles, six articles were included in the systematic review, and for meta-analysis, one study (that was not AOR) was excluded. The participants included in the studies were 2799 with the age range of 5–15 years old, and 93.6% were living in America. Four of the studies were placed in one group, due to having a common author (Perera). Moreover, a significant association was found between PAH exposure and ADHD in these studies (odds ratio = 2.57, 95% CI = 1.75–3.78); however, in all studies, there was no significant association between PAH exposure and ADHD for children (overall odds ratio = 1.99, 95% CI = 0.96–4.11) with low heterogeneity ($I^2 = 28.73\%$; P value < 0.001). This study provided a systematic review and meta-analytic evidence for the association between PAH exposure and ADHD by a small number of studies. Further research study can be conducted in various countries.

Keywords Attention deficit hyperactivity disorder · Children · Meta-analysis · Polycyclic aromatic hydrocarbon

Introduction

One of the main urban air pollutants is polycyclic aromatic hydrocarbons (PAHs) that fossil-fuel burning and other

combustion are the main sources of dispersion (Perera et al. 2012). PAHs are a large group of organic compounds enclosed with two or more benzene rings classified in various forms (Kim et al. 2013). The human exposure includes occupation;

Highlights

- Exposure to polycyclic aromatic hydrocarbons was related to an increased attention deficit hyperactivity disorder.
- There is a direct significant association between PAH exposure and ADHD (odds ratio = 2.57, 95% CI = 1.75–3.78).
- The total scale of DSM-IV (that is an instrument to screen for ADHD) in children with high maternal PAH metabolites is 3.3 times greater than children with low maternal PAH.

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passive and active smoking; and food, water, and ambient air pollution (Z. Li et al. 2008). Many studies have shown that PAH exposure, especially benzo[a]pyrene, increases the rate of multiple cancer types, including the lung, skin, bladder, respiratory, and urinary tract (Registry. 1995; Williams et al. 2013); neurodevelopmental outcomes (Genkinger et al. 2015); and neurological disorders (Wormley et al. 2004).

Some studies have concluded that socioeconomic factors, such as poverty and lack of education, as well as psychosocial factors, including maternal distress, are essential for child neurological disorders (Grote et al. 2010; Sandman et al. 2012; To et al. 2004). However, various research have shown the effect of prenatal PAH exposure on cognition (Peterson et al. 2015), on behavioral problems (Perera et al. 2018; Peterson et al. 2015), on IQ (Vishnevetsky et al. 2015), and on attention deficit hyperactivity disorder (ADHD) (Mortamais et al. 2017) in children.

Given the heightened susceptibility of the fetus during prenatal stages (Perera et al. 2014), and unique behavioral patterns, inhalation rates, body size, and higher ingestion of children, exposures to PAH and other environmental pollutants are of particular concern (Abid et al. 2014).

Laboratory studies have reported that PAH exposure during prenatal, neonatal, or adult stages causes behavioral effects, neurodevelopmental disorders (La’Nissa et al. 2007; Wormley et al. 2004), hyperactivity (Schellenberger et al. 2013), and symptoms of anxiety/depression and attention problems at ages 6–7 (Perera et al. 2012).

In the USA (US), approximately 11% of children with the age of 4–17 years old have been distinguished with ADHD (Visser et al. 2014). In some studies, ADHD affects approximately 5–7.1% of children and adolescents worldwide (Polanczyk et al. 2007; Willcutt 2012). ADHD is the most common behavioral (Bostic and P. J. 2008) and neurodevelopmental disorder diagnosed in children. Its main symptoms include impulsive hyperactive behavior, lethargic inattention, inability to focus on tasks, and functional impairment in family and academic settings (Barkley 2002) that are often accompanied with anxiety and depression (Spencer et al. 2007; Wilens et al. 2002). Neurodevelopmental and behavioral effects are identified with memory disorder and disability to learn (La’Nissa et al. 2007; Wormley et al. 2004) as well as anxiety- and depression-like symptoms in the absence of other overt toxicological effects (Saunders et al. 2006; Yokota et al. 2009).

Several studies have examined the relationship between PAH and ADHD. Given the existence of inconsistencies between them, the review of these studies is required.

However, one systematic review conducted by Aghayee et al. stated that the relationship between ambient gaseous and particulate air pollutants (PAH, dioxin, furans, and so on) and ADHD in children had been reviewed only qualitatively (Aghaei et al. 2019), but in the present study, it has been

examined and expressed as meta-analysis which in some studies have indirectly implied this relationship.

The basic question was “Is there an association between exposure to PAHs and the occurrence of ADHD in children?”

This study aimed to systematically explore and review all the published articles to investigate and identify the association between ADHD symptoms and PAH exposure during the prenatal and postnatal periods in children of nonsmoking women.

Methods

Search strategy

This systematic review and meta-analysis study was conducted in December 2018. Initially, the research question was: “Is there a correlation between exposure to PAHs and ADHD in children?” Accordingly, the main cause of ADHD in children is the exposure to PAHs of pregnant mothers, and one of the most important sources of PAH exposure in the nonsmoking general population is air. Articles on exposure to PAHs and ADHD in children of these mothers were searched. Moreover, the selected articles had checked the amount of PAHs in biological samples (urine and blood) in children with ADHD symptoms and their mothers. Therefore, a search strategy was formulated to completely identify the articles related to the main subject.

Searching the articles in this systematic review was performed in the databases of PubMed, Science Direct, Web of Science, Scopus, and Google Scholar. The search strategy included (“ADHD” OR “attention deficit hyperactivity disorder” OR “Neurological disorder”) AND (“Disorder, Attention Deficit-Hyperactivity” OR “Attention Deficit Disorders with Hyperactivity” OR “Attention Deficit Disorders with Hyperactivity” OR “Attention Deficit-Hyperactivity Disorders” OR “Attention Deficit-Hyperactivity Disorders”) AND (“PAHs” OR “polycyclic aromatic hydrocarbons” OR “polynuclear aromatic hydrocarbons” OR “phenanthrene” OR “pyrene” OR “naphthalene” OR “Benzo pyrenes” OR “Benz(a)Anthracenes”) AND (“Children” OR “primary school”) AND (“Environmental Pollution”) AND (“Environmental Exposure” OR “Maternal Exposure”) AND NOT (“smoking”).

Screening

Initially, the key strategy was searched in the mentioned databases and in the extracted relevant papers by investigating the titles and abstracts. Then, two reviewers critically and independently reviewed and evaluated all identified papers (the titles, abstract, and full text) for eligibility. This review was guided by a checklist assessing the perspicuity of the research.

Papers were assessed and listed according to the methodological quality. From this list, the published articles which evaluated the effect of PAH on ADHD were identified.

Quality assessment

Included studies in meta-analysis were assessed by the Newcastle-Ottawa Scale (NOS). Studies with a total score of 0–3, 4–6, and 7–9 were recognized as studies with poor, moderate, and high qualities, respectively (Table 1).

Selection criteria

With the following inclusion criteria, the articles were selected for this review: (a) children who were born from nonsmoking women; (b) prenatal exposure to PAH; (c) children that were exposed to environmental PAH; (d) child behavior was identified by a specific method; (e) evaluating exposure to PAHs; and (f) appropriate statistical analysis.

Some articles that studied the effects of combinations of PAH and other substances on ADHD in children, as well as a study that looked at the overlapping of Mrs. Perera's studies and results, PAH, and attention problems in children (Margolis et al. 2016) were excluded from this review.

Data extraction

Summarizing of selected articles was done by two reviewers using a data abstraction sheet. The abstracted information was sorted into categories based on exposure type, age of children, sample size, method of exposure assessment, country of study, and adjusted OR (95% CI) (Log OR) (Table 2).

Data analysis

After sorting the extracted information from the evaluated articles, meta-analysis on the studies' data was performed using Stata version 11.2. Metan order was used to analyze the adjusted odds ratio (log OR) (OR) with 95% confidence intervals (CIs) for the ADHD and PHA exposure, which were calculated. The heterogeneity analyzed by heterogeneity index (I^2) and the meta-regression (Higgins and Thompson 2003) tentatively suggest adjectives of low for I^2 values between 25 and 50%, moderate for 50–75%, and high for $\geq 75\%$. Finally, publication and small study bias was examined using Egger's test funnel plots. Analyses were conducted using Stata version 14.

Results

The comprehensive search identified 959 studies, followed by removing the duplicated studies ($n = 40$). The systematic

search that was carried out in the databases up to September 2018 produced an overall of 959 papers, 16 of which were from Web of Science, 163 from PubMed, 353 from Scopus, and 424 from Google Scholar.

A total of 11 papers were retrieved in full text after a primary screening based on the title ($n = 810$), on the abstract ($n = 97$), and on review articles and conference abstracts ($n = 2$). Out of these, five papers were excluded because they did not meet the inclusion and exclusion criteria including: combined effects of nutrition and PAH and ADHD, combined effects of prenatal exposure to PAH and material hardship on child ADHD behavior, neurodevelopment, and studies with incomplete data.

Consequently, a total of six articles were assessed to find the relationship between PAH exposure and ADHD (Fig. 1).

Table 2 shows the characteristics of studies entered in the meta-analysis and systematic review.

Methods Prospective cohort studies were the most frequent study type among the selected articles (Mortamais et al. 2017; Perera et al. 2014; Perera et al. 2012; Perera et al. 2011; Perera et al. 2018), and one study was cross-sectional (Abid et al. 2014).

Study population The total participants in the included studies were 2799 individuals, which in the meta-analysis, 2557 participants were included. The greatest number of participants was 1257 individuals at ages 6–15 years old, as reported in one cross-sectional study (Abid et al. 2014), whereas the lowest number of participants was reported as 242 individuals at ages 8–12 years old in this study (Mortamais et al. 2017). Most studies were performed in the USA (Abid et al. 2014; Perera et al. 2014; Perera et al. 2012; Perera et al. 2011; Perera et al. 2018), and one of the studies was conducted in Spain (Mortamais et al. 2017). Some studies or a number of studied groups revealed no significant relationship between PAH exposure and ADHD in children of nonsmoking women. But the overall results of the meta-analysis showed a significant positive relationship. Table 2 represents comprehensive summary findings of the studies included in this systematic review and meta-analysis. Given the fact that there was 2 to 3 adjusted odds ratio (AOR) in each article, a separate meta-analysis was conducted for each article by calculating a unit effect size and by completing the final meta-analysis.

Mortamais in 2017 (Mortamais et al. 2017) studied the effect of exposure to PAHs on basal ganglia and ADHD symptoms in primary schoolchildren in Spain. The results of this study showed that ADHD symptoms and inattentiveness were increased in children with higher exposure to BPA; however, a number of these associations were not statistically significant. Because it uses regression, it was inter-review systematic.

In the study by Perera (Perera et al. 2014), two types of exposure methods (maternal adduct and cord adduct) were

Table 1 Methodological quality of studies included in the meta-analysis

Cohort studies	Selection		Comparability		Outcome		Follow-up long enough for outcome to occur	Adequacy of follow-up of cohorts	Score
	Representativeness of the exposed cohort	Selection of the unexposed cohort	Ascertainment of exposure	Outcome of interest not present at start of study	Control for important factor or additional factor	Outcome assessment			
Perera; 2014; USA.	*	*	*	*	*	*	*	*	8
Perera; 2011; USA.	*	*	*	*	*	*	*	*	8
Perera; 2012; USA	*	*	*	*	*	*	*	*	8
Perera; 2018; USA	*	*	*	*	*	*	*	*	8

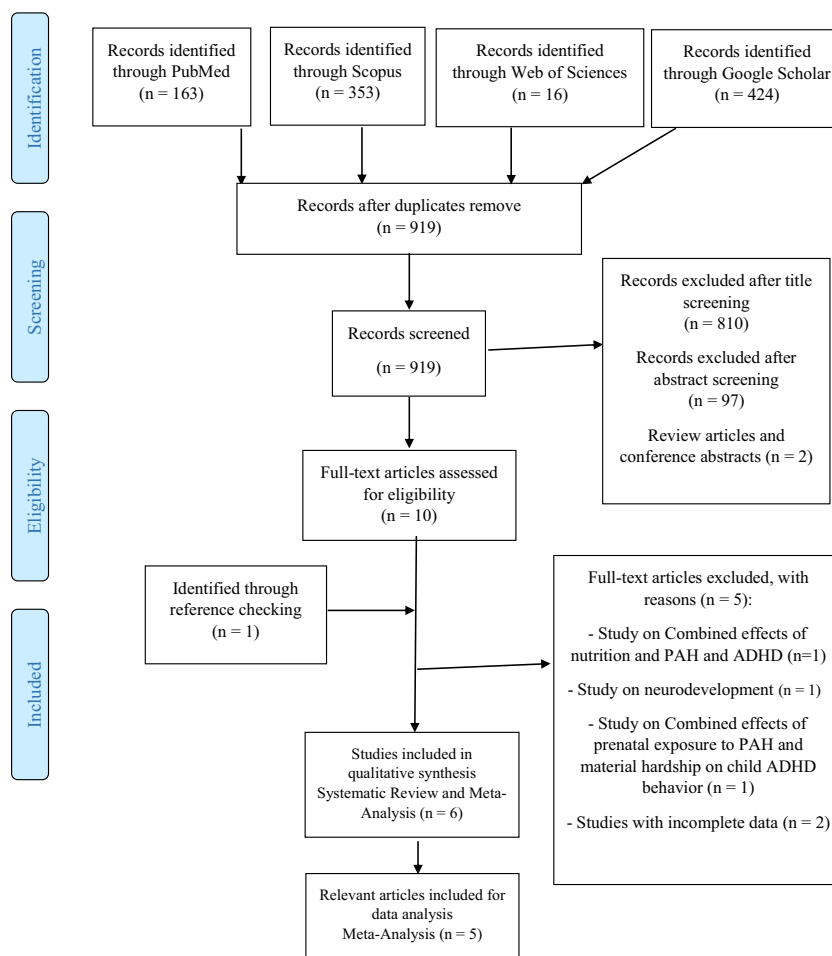
Cohort studies	Cross-sectional study	Selection		Comparability		Outcome		Score
		Representativeness of the sample	Non-respondents	Ascertainment of the exposure	Confounding factors are controlled	Outcome assessment	Statistical test	
Perera; 2014; USA.	Abid; 2014; USA.	*	*	*	*	*	*	8
Perera; 2011; USA.								
Perera; 2012; USA								
Perera; 2018; USA								

Table 2 Characteristics of studies entered in the meta-analysis and systematic review

Author; year; country	Study population	Study size	Age	Length of follow-up (year)	Measure exposure	Measure outcome	Adjusted OR (95% CI)	Adjustment for covariates
Perera; 2014; USA (15)	In this prospective cohort Children of nonsmoking African-American and Dominican women	233	9	9 (1998 and 2003 through the local prenatal care clinics into a prospective cohort study 98)	Maternal adduct Cord adduct	Attention deficit hyperactivity disorder behavior problems were assessed using the Child Behavior Checklist and the Conners Parent Rating Scale	1.48 (0.38–5.79) 1.17 (0.24, 5.66)	Postnatal PAH exposure, prenatal ETS, child sex, maternal education, child ethnicity, gestational age, maternal demoralization, heating season, HOME caretaking environment, maternal intelligence, child age at assessment, maternal ADHD, and child anxiety/depression at age 9
Perera; 2011; USA (29)	In this prospective cohort Children of nonsmoking African-American and Dominican women	96 205	5 7	1998 and 2003 through the local prenatal care clinics into a prospective cohort study	Cord adduct	CBCL syndrome scores: attention problems	5.66 (0.64–50.05) 3.30 (1.22–12.54)	Prenatal ETS, sex of child, gestational age, maternal IQ, HOME inventory, maternal education, ethnicity, prenatal demoralization, age at assessment, and heating season
Perera; 2012; USA (1)	Children of nonsmoking African-American and Dominican women in New York City (NYC)	253	6–7	1998 and 2003 through the local prenatal care clinics into a prospective cohort study	PAH Maternal adducts Cord adducts PAH Maternal adducts Cord adducts	CBCL: Attention problems CBCL: DSM-IV Attention	3.79 (1.14, 12.66) 2.24 (0.74, 6.77) 4.06 (0.99, 16.63) 2.3 (0.79, 6.70) 1.84 (0.66, 5.12)	Prenatal ETS, sex of child, gestational age, maternal IQ, HOME inventory, maternal education, ethnicity, prenatal demoralization, age at assessment, and heating season
Mortamais; 2017; Spain (14)	The Brain Development and Air Pollution Ultrafine Particles in School Children (BREATH) project (Saunders et al. 2006)	242	8–12		Outdoor PAH Indoor PAH	ADHD symptoms (ADHD/Diagnostic and Statistical Manual of Mental Disorders)	RR 1.02 (0.81, 1.29) RR 0.94 (0.85, 1.04) rate ratio (RR)	Age, sex, maternal education, and residential neighborhood socioeconomic status
Abid; 2014; USA (16)	National Health and Nutrition Examination Survey (NHANES) 2001–2004	1257	6–15	Cross-sectional	Urinary metabolite concentrations	Parental report of (1) ever doctor-diagnosed ADHD	0.91 (0.64–1.51) (mean)	Age, race/ethnicity, sex, creatinine, smokers in the household, PIR, birth weight, and having a routine source of medical care
Perera; 2018; USA (12)	The Columbia Center for Children's Environmental Health (CCCEH) cohort study	351	9	Cohort		The Conners Adult ADHD Rating Scales (CAARS)	4.05 (1.42–11.7)	Gender, ethnicity, prenatal ETS, maternal education, gestational age, TONI, HOME, age at assessment, maternal ADHD, child anxiety/depression at age 9, and heating season

PAHs Polycyclic aromatic hydrocarbons, ADHD Attention deficit hyperactivity disorder, CI Confidence interval, HF High frequency, LF Low frequency, OR Odds ratio, SD Standard deviation, AOR Adjusted OR

Fig. 1 Flowchart of the database search, selection, and review process of articles



investigated. AOR was associated in maternal adduct exposure (AOR = 1.48 (95% CI 0.38 to 5.79)) and cord adduct exposure (AOR = 1.17 (95% CI (0.24, 5.66))). These two methods were combined to get the mean of 1.34 which is appropriate for meta-analysis.

In the second study by Perera et al. (2011), PAH adducts in cord blood cells were measured as a unite exposure; then, associations between PAH and behavior outcomes at ages 4.8 and 7 years old were reported by two separate checklists (CBCL and DSM); CBCL AOR 5.66 (0.64–50.05) and 3.30 (1.22–12.54) in DSM. Moreover, in this study, the two AORs were combined, and the mean (4.19) was obtained until it enters the meta-analysis.

In the study of behavior at age 6–7 years old by Perera et al. (2012), three types of PAH exposure were monitored which are PAH of personal air concentrations, maternal, and cord blood PAH exposures. In the next step, the associations between all the above measures and outcomes in children were compared by two separate checklists (CBCL and DSM). CBCL AORs were 3.79 (95% CI (1.14,12.66)), 2.24 (95% CI (0.74, 6.77)), and 4.06 (95% CI (0.99, 16.63)), and DSM AORs were 2.3 (95% CI (0.79, 6.70)), 1.84 (95% CI (0.66,

5.12)), and 2.64 (95% CI (0.68, 10.26)) in PAH exposure, maternal, and cord PAH, respectively.

At first, AORs related to exposure methods were combined (air, maternal, and cord PAHs) by CBCL and DSM checklists. The means of AORs related to air, maternal, and cord PAHs were 2.87, 2.02, and 3.24, respectively. Finally, 3 above AORs were combined, and the ultimate AOR was calculated as 2.55.

Abid et al. studied urinary polycyclic aromatic hydrocarbon metabolites and ADHD on 1257 American children 6–15 years old (Abid et al. 2014). In this study, metabolite concentrations of eight PAHs were included: 1-pyrene, 1-naphthol, 2-naphthol, 2-fluorene, 3-fluorene, 1-phenanthrene, 2-phenanthrene, and 3-phenanthrene. Then, the relationship between PAH metabolites (log-transformed and high-versus-low) and ADHD was measured from a statistical point reported. In total, AOR log-transformed and high-versus-low metabolites were 0.91 (95% CI (0.83, 1.00)) and 0.91 (95% CI (0.7, 1.18)), respectively.

The last study entered in the meta-analysis was conducted in 2018 by Perera. This study measured the biomarker of prenatal PAH exposure, PAH-DNA adducts in maternal blood

at delivery, and maternal material hardship among 351 children aged 9 (Perera et al. 2018). Children with high prenatal PAH exposure (high adducts) generally had more symptoms of ADHD (higher scores) compared with those with low PAH exposure of AOR 4.05 (95% CI (1.42–11.7)) (significant positive association).

Among 6 papers, 5 papers entered the meta-analysis (Abid et al. 2014; Perera et al. 2014; Perera et al. 2012; Perera et al. 2011; Perera et al. 2018), where 4 of which were from a cohort study. The OR of these studies was combined, and the final AOR was 2.5 (95% CI (1.75–3.7)).

In the meta-analysis of PAH on ADHD for all studies, including studies of Frederica P. Perera (4 studies) and Zaynah Abid (1 study) by the random-effect model, it showed that PAH exposure has no significant association with ADHD (AOR = 1.99 (95% CI 0.96–4.11)) with significant heterogeneity ($I^2 = 28.73\%$; P value < 0.001), so this study has a low heterogeneity (Fig. 2). Then, with regard to that, the number of studies included in meta-analysis was very small (5 studies); the meta-regression was used to identify heterogeneity by study years and study type. Initially, we assessed the study year, which was nonsignificant (P value = 0.926). Then, the study type was assessed, which was significant (P value = 0.015) (Table 3). According to these results, next, we conducted a sensitivity analysis by 4 studies of Perera that were cohort studies. The results showed that the PAH exposure has a significant association with ADHD (AOR = 2.57 (95% CI 1.75–3.78)) with nonsignificant heterogeneity ($I^2 = < 0.01\%$; P value = 0.443) (Fig. 3).

Finally, evidence of publication bias was not detected using Egger's test ($P = 0.061$), and a funnel plot symmetry was observed in the meta-analysis (Fig. 4). The results remained unaltered after the trim and fill analysis (OR fixed = 0.96, 95% CI 0.88–1.05; OR random = 1.98, 95% CI 0.96–4.10), suggesting stable results.

Fig. 2 Odds ratio and 95% CI of the association between exposure to polycyclic aromatic hydrocarbons (PAH) and attention deficit hyperactivity disorder (ADHD) in the children. Random-effect model was used to calculate the overall OR and 95% CI for all studies

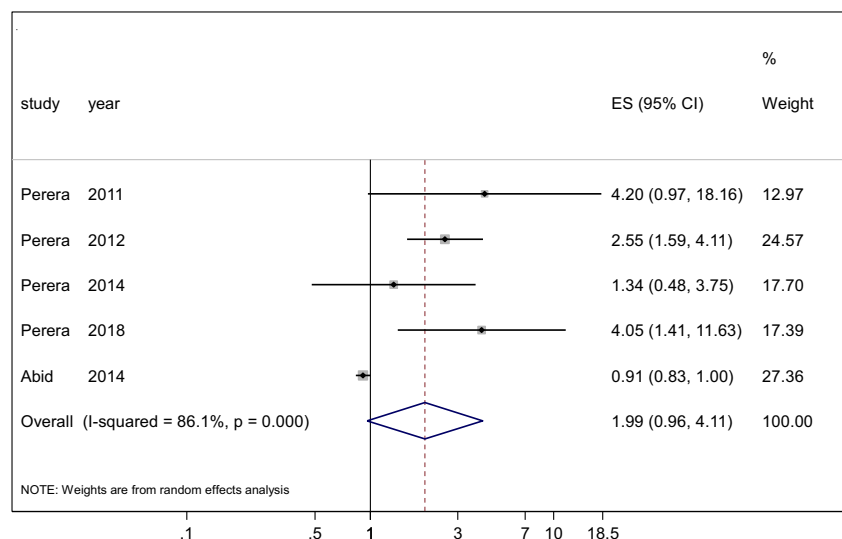


Table 3 Meta-regression analysis of sources of potential diagnostic heterogeneity on the association of PAH exposure and ADHD

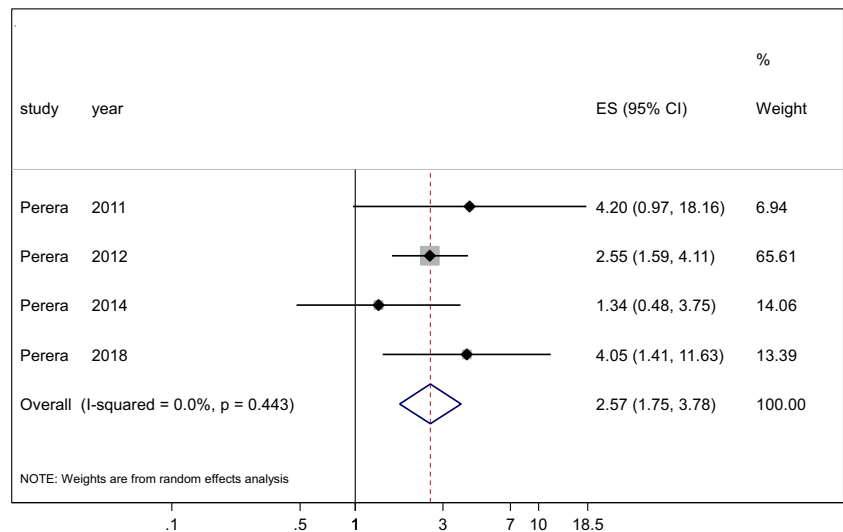
Predictor variable	Beta	SE	T	P value
Step 1				
Study year	0.016	0.163	0.10	0.926
Intercept	− 32.684	328.99	− 0.10	0.927
Step 2				
Study type	− 1.032	0.202	− 5.10	0.015
Intercept	0.942	0.196	4.79	0.017

Discussion

In this systematic review and meta-analysis, the association of the ADHD and PAH exposure was assessed in children of nonsmoking women. This study was conducted by different tests to assess the relationship between PAH and ADHD.

Numerous factors increase ADHD in children, such as PAH exposure. In a study conducted by Perera P et al., it was demonstrated that ADHD symptoms in children with high prenatal PAH exposure had a significant increase compared with those children with low exposure (Perera et al. 2018). Furthermore, some studies have pointed out the relationship between ADHD and behavioral problems with PAH exposure (Perera et al. 2013; Perera et al. 2011). Some studies have indirectly implied this relationship (Genkinger et al. 2015; Perera et al. 2006; Peterson et al. 2015). In nonsmoking individuals, more than 70% of the total exposure to PAH was caused by many indoor combustion sources (cooking, incense, etc.) and dietary consumption (Khillare et al. 2012; Martorell et al. 2011; Wang et al. 2018). However, in studies that focused on the association of the ADHD and PAH exposure, the exposure was checked through the air (Abid et al. 2014; Mortamais et al. 2017; Perera et al. 2012). Some

Fig. 3 Odds ratio and 95% CI of the association between exposure to polycyclic aromatic hydrocarbons (PAH) and attention deficit hyperactivity disorder (ADHD) in the children. Random-effect model was used to calculate the overall OR and 95% CI for all cohort studies



references have indicated that the evidence for the association of environmental risk factors and ADHD is insufficient (Lam et al. 2017; Suades-Gonzalez et al. 2015). However, the results of the study by Saez et al. showed that being in the range of environmental factors could be correlated with ADHD incidence (Saez et al. 2018). Nevertheless, in order to prevent the publication bias in the articles that have positive results, careful interpretation is required. Therefore, the study and review by several experts play an important role in minimizing publication bias.

The metabolites are known as indicators of constant exposure in populations for ambient PAH monitoring (Zhao et al. 1990). The study of Perera showed that with high maternal PAH metabolites, total scales of DSM-IV (that is an instrument to screen for ADHD) (Achenbach and Rescorla 2001; Conners 2001) were 3.3 times greater than children with low maternal PAH (Perera et al. 2014); however, the results were not statistically significant. In this study, the ADHD behavioral problems were increased

by prenatal exposure; however, there was no significant correlation between prenatal exposure monitored in PAH and ADHD (Perera et al. 2014).

Benzo[a]pyrene is a carcinogenic member of the class of PAHs (Leavitt et al. 2008) negatively affecting the brain function (McCallister et al. 2008). Mortamais et al. demonstrated that with high exposure to benzo[a]pyrene in children, ADHD symptoms increased; however, the associations were not statistically significant (Mortamais et al. 2017).

The results of this study showed that by increasing BAP levels from indoor to outdoor, ADHD scores were increased (Peterson et al. 2015) that were consistent with the study of Perera F (Perera et al. 2012). It was proved that exposure to environmental PAH of NYC air has a negative effect on children's behavior (Perera et al. 2012).

Phenanthrene is a member of the class of PAHs that has more plentiful metabolites in urine (Jacob and Seidel 2002). Abid et al. studied the association of phenanthrene metabolites and ADHD. The study showed that children exposed to high phenanthrene metabolite levels were not significantly associated with ADHD (Abid et al. 2014). This difference in the results may be due to diagnostic report.

In studies of Perera et al., it was observed that the OR between PAH exposure and CBCL scores was decreased by aging (Mortamais et al. 2017; Perera et al. 2014; Perera et al. 2012; Perera et al. 2011) that is in line with the study of Abid et al. in children aged 6 to 15 (Abid et al. 2014). The possible reason that several studies have emphasized them is the timely diagnosis and treatment of ADHD (Halldner et al. 2014; Morrow et al. 2012; Zoëga et al. 2012). On the other hand, some studies believe that the first diagnosis of ADHD in schoolchildren happened by teachers and other school staff (Phillips 2006) due to their tendency to feel insecure in confronting behavioral problems (Walter et al. 2006).

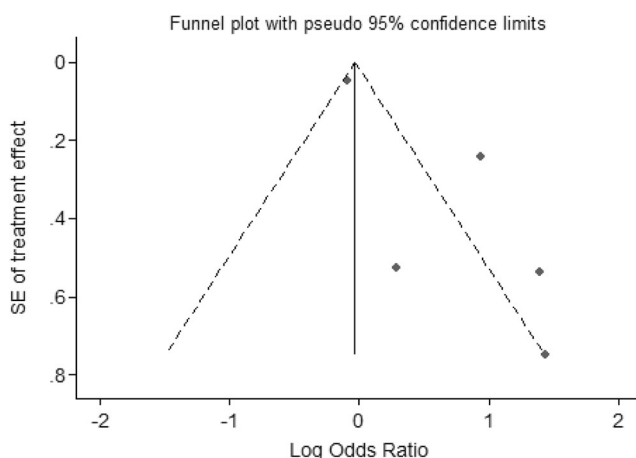


Fig. 4 Funnel plot for publication bias analysis results

The study of the effects of PAH on brain structure by Bradley et al. proved that the white matter surface was decreased by increasing prenatal PAH exposure that slows down the speed of information processing and incidence of ADHD symptoms (Peterson et al. 2015). The study of Ming-guo Qiu et al. showed that structural changes in the brain affected ADHD (Qiu et al. 2011; Spencer et al. 2013), so that the volume of the white matter, cerebral cortex, and caudate nucleus decreased in ADHD patients (Spencer et al. 2013). Moreover, the results of Marion Mortamais' study showed that by increasing the ADHD score, caudate and putamen nucleus (Mortamais et al. 2017) and CNV decreased in children with ADHD (Frodl and Skokauskas 2012). Some studies have demonstrated the destructive effects of benzo[a]pyrene on gene expression and structure change in nerve cells, as one of the possible reasons of the previous studies (Chepelev et al. 2015; Li and Tsien 2009).

Given that the included studies were heterogeneous, their results were investigated using a random-effect model; moreover, in the overall analysis, the significant heterogeneity was observed ($I^2 = 28.73$). The factors creating heterogeneity in this meta-analysis were included (study type and sample size). In this study, several limitations were observed, including inadequate studies, especially cohort studies in different populations. The publication bias was an important feasible limitation which any attempt to obtain a reliable joint estimation of effects was distorted to significant results (Thornton and Lee 2000). Therefore, the results should be interpreted with caution and by several experts.

Conclusion

The results of this meta-analysis demonstrated a direct association between PAH and ADHD. This association was a significant positive association. Therefore, according to this study, it was concluded that PAH has a vital role in child ADHD.

The results of this systematic review can also provide useful information for decision-making in engineering practices in this domain.

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